

Project Overview

US County Income Prediction

Built a clean, county-level dataset and modeled **how economic and social factors relate to mean household income** across the U.S. We integrated Census, BLS, and Zillow data via FIPS codes, handled missingness with geographically informed K-NN imputation, and compared linear vs. tree-based regressors.

Objective: Explain and predict county-level mean household income from economic and social indicators.

Approach: Collect data via APIs/CSVs, normalize and merge on county basis, then visualize patterns and train regression models.

Validation: Train/hold-out evaluation with standard regression metrics; model choice guided by comparative performance and interpretability.



Scope

~3,000 counties
3+ sources
15+ features



Stack

Python
Different Models
K-NN imputation



Datasets

US Census
BLS
Zillow

Architecture

Data Sources



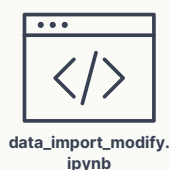
US Census API



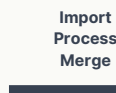
BLS



...



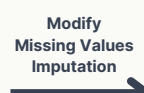
data_import_modify.
ipynb



Import
Process
Merge



data/
merged_county_data.csv



Modify
Missing Values
Imputation



project.ipynb



Visualizations



Modeling



[Github repository](#)

Results

Integrated dataset

Extensive demographic and economic data from different sources combined into one unified dataset.

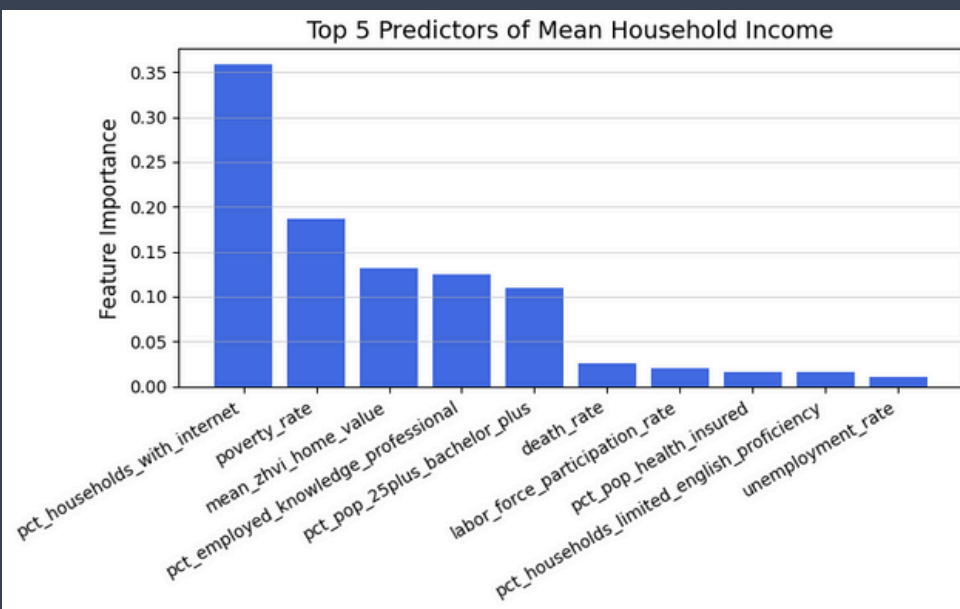
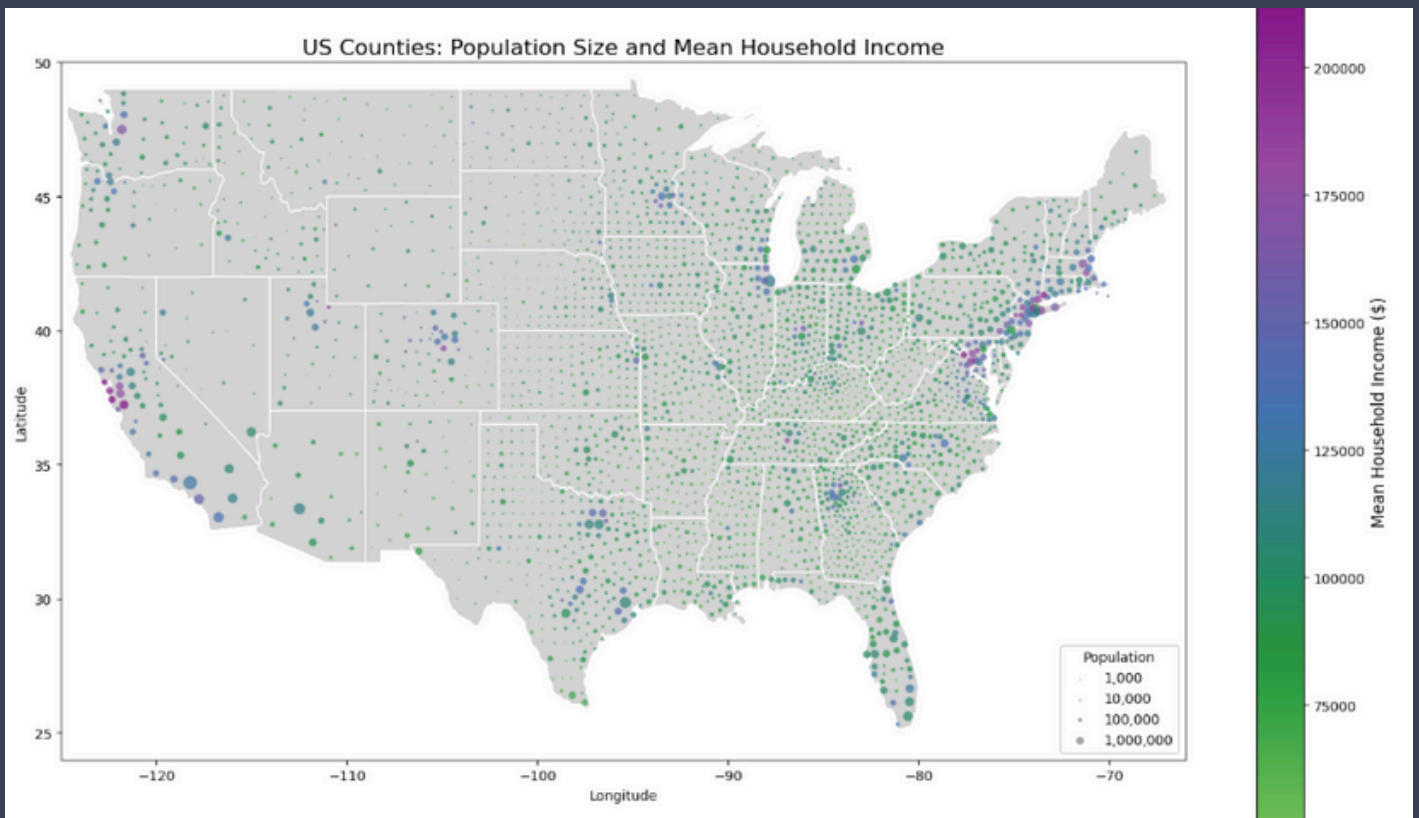
Complete Coverage

Eliminated missing values via geo K-NN imputation for more robust forecasts and results.

Modeling suite

Comparison of multiple different models and approaches for maximum accuracy.

Visualizations / Modelling



With an R2 of 0.86 Random Forest performed the best.

It predicts the mean income based on different variables, with some being of greater importance (figure left)